Final Project Guide

Module 7: Advanced Machine Learning & AI

The Module 7 Final Project is your opportunity to apply advanced machine learning and deep learning techniques to a real-world dataset. You will work in teams on a problem, explore the data, train and refine a deep learning model, and communicate your findings both technically and professionally. Your goal is to tell a compelling, data-driven story supported by deep learning.

Complete instructions will be available in the week in which each component is assigned.

The project has five components:

Component	Due	Points
Team Contract	Week 7	5
Project Milestone 1	Week 8	25
Project Milestone 2	Week 11	100
Final Project Report	Week 13	180
Team Member Evaluation	Week 14	10

Tips:

- Work collaboratively, divide tasks fairly, and communicate frequently.
- Ask for feedback and guidance during office hours and Live Sessions.
- Plan as a team and get started early — don’t wait until the due dates!

Team Contract [5 pts]

The first step in the final project is to connect with your team and discuss how you will work together. The team contract is an agreement between all team members about how you will collaborate to complete the final project. It invites you to determine together when and how often you will meet, how you will communicate, and how quickly everyone should respond to messages. The contract also asks everyone to agree to do their fair share of the work and finish tasks on time. A thoughtful approach to adopting this contract will prevent confusion, keep everyone accountable, and make it easier to work as a team to finish the project successfully.

Teamwork Resources

Project Milestone 1: Task and Dataset Selection & Problem Framing [25 pts]

In this milestone, we will provide you with a selection of five tasks and datasets. Your team will choose one task and dataset that you would like to work with for the final project.

Follow these steps adapted from: Geron, A. (2023). *Hands-on machine learning*. O'Reilly Media.

Part 1: Choose the Dataset [10 pts]

- Select a task and dataset
- Perform preliminary exploratory data analysis (EDA)

Datasets will be available in the Milestone 1 notebook.

Part 2: Frame the Problem [15 pts]

- Define the prediction task clearly (classification or regression).
- Identify input and output formats.
- Specify success metrics (e.g., accuracy, F1, mean absolute error (MAE), root mean squared error (RMSE)).

Deliverable:

You will submit a single Jupyter notebook as a group submission via your team leader's Gradescope account, including:

1. Dataset description
2. Problem framing
3. Initial EDA (with plots or tables)

You are expected to document your use of artificial intelligence (AI) in accordance with Faculty of Computing & Data Sciences (CDS) policy in the syllabus. We reserve the right to adjust grades if the CDS AI policy is not followed.

Project Milestone 2: Data Preparation and Building the First Deep Model [100 pts]

The goal of Milestone 2 is to prepare your data for modeling and build your first working deep learning prototype. This replaces the classical workflow that you learned in Module 3, of heavy preprocessing and feature engineering, which emphasized modeling, representation learning, and training strategy.

Follow these steps adapted from Geron's *Hands-On Machine Learning*:

Part 1: Data Preparation and Initial Exploration [50 pts]

- Download or load the dataset.
- Conduct minimal preprocessing needed for deep models.
- Visualize data distributions and class balance.
- Standardize all features (not the target) in preparation for modeling.

Part 2: Build and Evaluate First Version of the Deep Model Prototype [50 pts]

- Build and train a baseline deep learning model (feedforward neural network (FFNN), convolutional neural network (CNN), recurrent neural network (RNN), GPT, etc., depending on your data).
- Evaluate performance on a validation set.
- Include auxiliary layers (batch normalization, dropout, etc.) as needed to access baseline performance (you'll fine-tune further for the final report).

Deliverable:
You will submit a single Jupyter notebook as a group submission via your team leader's Gradescope account, including:

- Clear description of the problem and dataset.
- EDA and preprocessing.
- Code and explanation of your first working deep model.
- Initial training and validation performance.

You are expected to document your use of AI in accordance with CDS policy in the syllabus. We reserve the right to adjust grades if the CDS AI policy is not followed.

Final Project Code + Report [180 pts]

This is your **final submission**, integrating code, results, and insights.

Follow these steps adapted from Geron's *Hands-On Machine Learning*:

Part A: Final Code Repository [80 pts]

- Select your best deep learning model based on validation/test performance.
- Retrain it cleanly from scratch, using the full training set.
- Document your final architecture and training process, including:
  - Use of regularization (e.g., dropout, weight decay).
  - Hyperparameter tuning (optimizer, learning rate, batch size, epochs).
  - Use of callbacks (early stopping, learning rate scheduler).
- Include charts/plots: training versus validation loss, accuracy, confusion matrix, etc.
- Make sure code is clean, readable, and well-commented.

This section serves as your project's "code deliverable" — a reusable and well-documented notebook suitable for continued development.

Part B: Technical Report [100 pts]
Write a professional, structured report communicating your project goals, methodology, and findings.

Follow a structure similar to:

- CMU guide to structure
- Data science report example

Audience: Data science colleagues who are technically literate but unfamiliar with your specific project.

Include:

- **Problem definition** — what you are solving and why it matters
- **Data overview** — source, features, and key characteristics
- **Modeling approach** — description of your modeling choices and rationale
- **Training strategy** — Loss functions, optimizers, learning rates, etc.
- **Evaluation** — metrics, visualizations, interpretation
- **Key results and analysis**
- **Limitations and future work** — what didn't work, what's next

Do not assume the reader will read Part 1. Include only essential code excerpts, not full notebooks. Prioritize clarity and insight.

The technical report should be between 5–10 pages double-spaced in PDF format.

Deliverable

You will submit both Part A (code) and Part B (report) as a single group submission via your team leader's Gradescope account.

You are expected to document your use of AI in accordance with CDS policy in the syllabus. We reserve the right to adjust grades if the CDS AI policy is not followed.

Final Project Team Member Assessment [10 pts]

The Team Member evaluation asks you to provide feedback to each of your team members about their role in the completion of the Final Project. The goal of feedback is to help everyone improve their teaming skills by reflecting on their own actions and how they've impacted the team's performance.

The evaluation has three steps:

1. Use Plus Delta to evaluate yourself and each of your team members
2. Read received reviews
3. Write a self-reflection on the feedback you received